Alex — 37/40 (Q1 12/12 · Q2 11/13 · Q3 14/15)

南京外国语学校
NANJING FOREIGN LANGUAGE SCHOOL
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CIE Semester 1 Midterm Examination 2026 - 2027

Paper 1&2 Combined

AS Computer science

60 Minutes

No additional materials are needed

READ THESE INSTRUCTIONS FIRST

Write in dark blue or black pen.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use red ink, staples, paper clips, highlighters, glue or correction fluid.
The exam paper contains 3 structured questions.
Answer **all** your questions on the **exam booklet.**

Q1	**/ 12**
Q2	**/ 13**
Q3	**/ 15**
Total	**/ 40**

The number of marks is given in brackets [ ] at the end of each question or part question.

This document consists of 8 printed pages.

Cambridge Assessment
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[TURN OVER]

1. A gym uses a program to store information about its members.

(a) State the most appropriate **data type** for each item of data.

Item of data	Data type
the membership number — a whole number between 1 and 9999	~~Integer~~ INTEGER
the monthly fee — for example 299.50	~~Real~~ REAL
whether the member has paid for this month	~~Boolean~~ BOOLEAN
the membership grade — a single letter, A, B, C or D	~~Char~~ CHAR

[4]

(b) The following pseudocode is run.

```
DECLARE Count : INTEGER
DECLARE Total : INTEGER

Count <- 1
Total <- 0

WHILE Count <= 4
   Total <- Total + Count * 2
   Count <- Count + 1
ENDWHILE

OUTPUT Total
OUTPUT Count
```

Complete the trace table below.

Count	Count <= 4 ?	Total
1	TRUE	2
2	TRUE	6
3	TRUE	12
4	TRUE	20
5	FALSE	

Output:

Total = 20

Count = 5 [4]

(c) A programmer writes the following pseudocode. It is intended to output the numbers 10, 9, 8, 7, 6, 5, 4, 3, 2, 1.

```
DECLARE N : INTEGER

N <- 10

WHILE N >= 0
   OUTPUT N
   N <- N + 1
ENDWHILE
```

(i) State the error in this pseudocode and explain what happens when it is run.

The whole loop will not end as N will never be less then 0 when it is run. And the programmer wants to output numbers from 10 to 1, so N←N+1 is wrong, which is a logic error. It will output 0 as well. [2]

(ii) Rewrite the pseudocode so that it outputs the numbers as intended.

```
DECLARE N : INTEGER
N←10
WHILE N>0
   OUTPUT N
   N←N-1
ENDWHILE
```

[2]

2. The following pseudocode defines a function.

```
FUNCTION Calc(A : INTEGER, B : INTEGER) RETURNS INTEGER
   IF A > B THEN
      RETURN A * 2
   ELSE
      RETURN B + A
   ENDIF
ENDFUNCTION
```

(a) State the value returned by each of the following calls.

Call	Value returned
`Calc(7, 3)`	14
`Calc(2, 9)`	11
`Calc(5, 5)`	10

[3]

(b) Write pseudocode for a function `DigitTotal` that takes an `INTEGER` parameter and **returns** the sum of its digits. For example, `DigitTotal(472)` returns 13.

```
FUNCTION DigitTotal(Num:INTEGER) RETURNS INTEGER
DECLARE Sum, i : INTEGER
Sum←0
FOR i ← 1 to Length(Num)
Sum ← Sum + MID(Num, i, 1)
NEXT i
RETURN Sum
ENDFUNCTION
```

[5]

Q2(b) 3/5
The cause is right, but the task expects N MOD 10 / N DIV 10 to take each digit. You attempted the string route, and even then the number is never converted: LENGTH(Num) and MID(Num, i, 1) do not work on an INTEGER. Convert first: str <- NUM_TO_STR(Num), then LENGTH(str) and STR_TO_NUM(MID(str, i, 1)). (-2)

(c) Explain the difference between a **function** and a **procedure**.

Function could return a value and the function will stop running as long as a value is returned.
Procedure will not return a value but it could adjust the existed variables through BY REF

[2]

(d) Explain what is meant by **modularization** and give **one** benefit of using it.

Modularization is uses a function or a procedure to represents plenty of codes. When calling them, they will run the codes and give a feedback. Using modularization coud simplify the length of the code and be easier for programmer to check or read. [3]

3. A school stores the scores of 30 students in a one-dimensional array `Scores`.

(a) Write pseudocode to declare the array `Scores`.

```
DECLARE Scores : ARRAY[1:30] of INTEGER
```

[2]

(b) The array `Scores` has been filled with 30 values. Write pseudocode to output the **largest** value in `Scores` and the **index** at which it is found.

```
DECLARE c,i, Max : INTEGER
Max ← Scores[1]
c ← 0
FOR i ← 1 to 30
IF Scores[i] > Max THEN
Max ← Scores[i]
c ← i
ENDIF
NEXT i
OUTPUT "the largest value :" & Max
OUTPUT "the index :" & c
```

[5]

Q3(b) 4/5

Correct except the starting position: c is initialised to 0, but the first element's index is 1. If Scores[1] is the largest, your program reports index 0. Set c <- 1 (and start the loop at 2), or initialise both values from the first element.

(c) Write pseudocode for a function `CountPass` that takes the array `Scores` and an integer `PassMark` as parameters and **returns** the number of values in `Scores` that are **greater than or equal to** `PassMark`.

```
FUNCTION CountPass (Scores : ARRAY[1:30] OF INTEGER, PassMark : INTEGER) RETURNS INTEGER
    DECLARE i, Count : INTEGER
    Count ← 0
    FOR i ← 1 TO 30
        IF Scores[i] >= PassMark THEN
            Count ← Count + 1
        ENDIF
    NEXT i
    RETURN Count
ENDFUNCTION
```

[5]

(d) The school also stores the marks for three subjects in a two-dimensional array `Marks`. `Marks` has **5 rows** and **3 columns**.

(i) Write pseudocode to declare the array `Marks`.

```
DECLARE Marks : ARRAY[1:5, 1:3] OF INTEGER
```

[1]

(ii) Write pseudocode to output **every** value in `Marks`.

```
DECLARE i : INTEGER
FOR i ← 1 TO 5
    OUTPUT Marks[i, 1]
    OUTPUT Marks[i, 2]
    OUTPUT Marks[i, 3]
NEXT i
```

[2]

```
DECLARE i, j : INTEGER
FOR i ← 1 TO 5
    FOR j ← 1 TO 3
        OUTPUT Marks[i, j]
    NEXT j
NEXT i
```

Q3(d)(ii) 2/2
The nested-loop version written on the right is correct - but the answer in the space uses one loop with three hard-coded OUTPUTs. Write the nested-loop answer in the answer space.

End of the exam.
Use following pages for scratch paper.